

1.1.5 Handwriting vs IA

Evidencia grafica de práctica.

- Preparación de entorno.

```
D:\>mkdir actividad_mnist

D:\>cd actividad_mnist

D:\actividad_mnist>py -3.11 -m venv .venv

D:\actividad_mnist>.venv\Scripts\activate

(.venv) D:\actividad_mnist>python --version
Python 3.11.9

(.venv) D:\actividad_mnist>mkdir D:\temp

(.venv) D:\actividad_mnist>set TEMP=D:\temp

(.venv) D:\actividad_mnist>set TMP=D:\temp

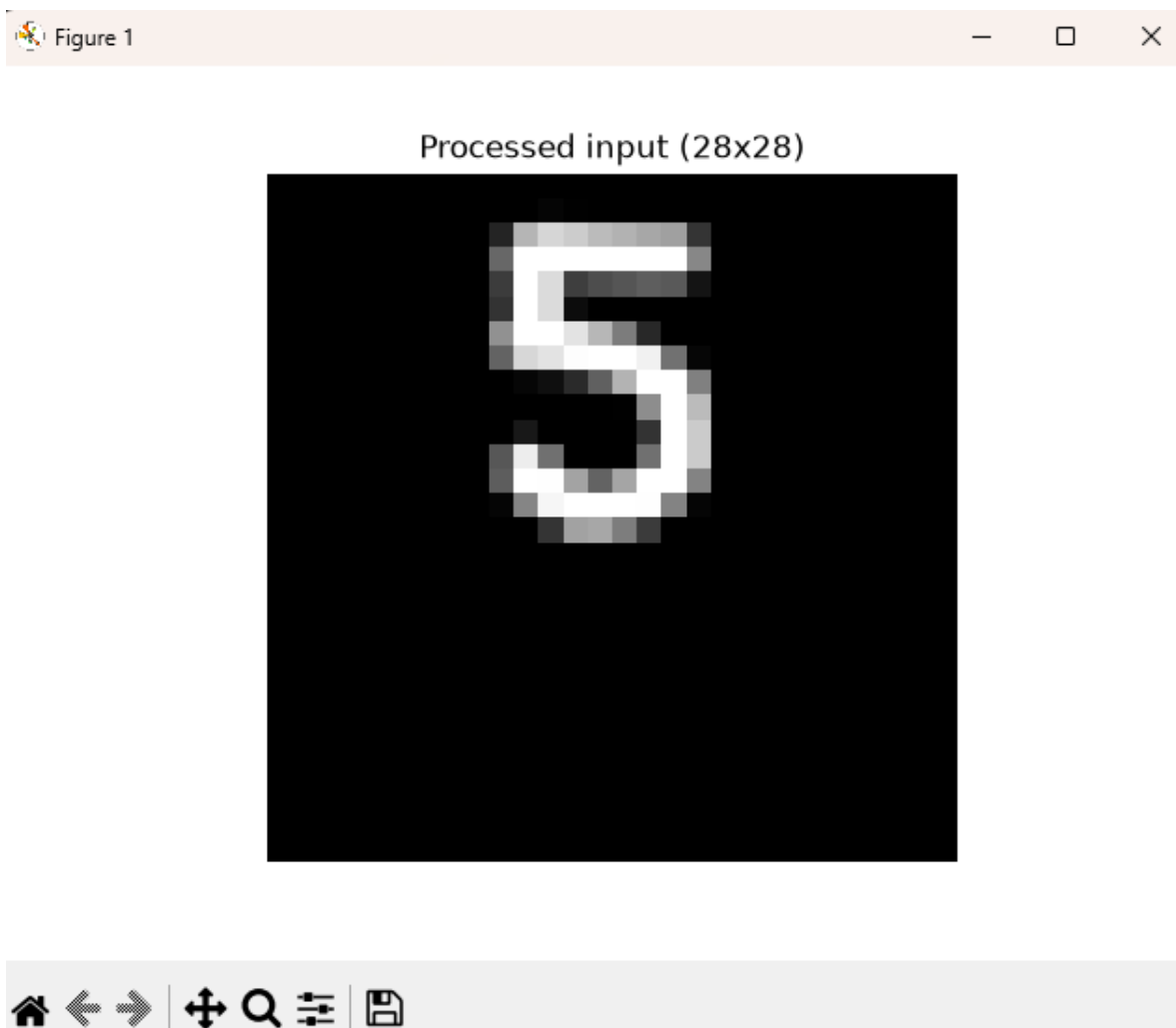
(.venv) D:\actividad_mnist>echo %TEMP%
D:\temp

(.venv) D:\actividad_mnist>python -m pip install --upgrade pip
Requirement already satisfied: pip in d:\actividad_mnist\.venv\lib\site-pa
ckages (24.0)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
  Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 791.6 kB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
```

- Resultados obtenidos.

```
(.venv) PS D:\actividad_mnist> & d:\actividad_mnist\.venv\Scripts\python.exe d:/actividad_mnist/test_letra.py
object as the first layer in the model instead.
  super().__init__(**kwargs)
I0000 00:00:1787728797.713860 19128 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX512F AVX512_VNNI FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
1/1 _____ 0s 180ms/step
1/1 _____ 0s 103ms/step
1/1 _____ 0s 229ms/step
MLP prediction: 3
CNN prediction: 3
RNN prediction: 5
(.venv) PS D:\actividad_mnist>
```

- Imagen analizada.



- Archivos.

